

3.1.2 Limitation of physical measurements

Content	Opportunities for skills development
Random and systematic errors. Precision, repeatability, reproducibility, resolution and accuracy. Uncertainty: Absolute, fractional and percentage uncertainties represent uncertainty in the final answer for a quantity. Combination of absolute and percentage uncertainties. Represent uncertainty in a data point on a graph using error bars. Determine the uncertainties in the gradient and intercept of a straight-line graph. Individual points on the graph may or may not have associated error bars.	PS 2.3 Students should be able to identify random and systematic errors and suggest ways to reduce or remove them. PS 3.3 Students should understand the link between the number of significant figures in the value of a quantity and its associated uncertainty. MS 1.5 Students should be able to combine uncertainties in cases where the measurements that give rise to the uncertainties are added, subtracted, multiplied, divided, or raised to powers. Combinations involving trigonometric or logarithmic functions will not be required.